

NLP and Reinforcement Learning-Based Chatbots for Efficient Public Service Delivery in Smart Cities

Sowmya V¹, Syed Fakruddin Albeez², Venkadesh Ramalingam³, G. Nagappan⁴, G.Manikandan⁵, Saranya R⁶

¹Department of computer science and applications, SRM Institute of Science and Technology, Ramapuram, Chennai.

²Department of Computer science, Desh Bhagat University, Mandi Gobindgarh, Punjab, India

³Department of Information Technology, College of Computing and Information Sciences, University of Technology and Applied Sciences - Shinas, Oman,

⁴Department of Computer Science Engineering, Saveetha Engineering College, Chennai, Tamil Nadu, India

⁵Department of Electronics and Communication Engineering, Saveetha School of Engineering, Saveetha Institute of Medical and Technical Sciences, SIMATS, Chennai, Tamilnadu, India

⁶Department of English, KPR Institute of Engineering and Technology, Coimbatore, Tamil Nadu, India

Emailid:sowmydev@gmail.com, albeez@hotmail.com, venkadesh.ramalingam@utas.edu.om, nagappan@saveetha.ac.in, manikandang.sse@saveetha.com, saranya.r@kpriet.ac.in

Abstract-Artificial intelligence (AI)-powered chatbots are revolutionizing public service delivery in Indian municipalities by increasing operational efficiency and citizen engagement. Physical visits and phone calls were instances of conventional approaches that were costly, time-consuming, and had little penetration. The AI-based chatbot system can automatically respond in real-time to citizens' questions, complaints, and service status checks because it is based on natural language processing (NLP) and machine learning (ML). Among its attributes are its round-the-clock accessibility, simple backend integration, and reinforcement learning for continuous learning. Performance evaluation shows reaction times have decreased from 5 to 20 minutes to 1 to 5 seconds, resolution rates have improved by 10 to 20%, and average user satisfaction has increased from 3.2 to 4.7 out of 5. With the extra benefit of being cost-effective, the proposed method improves productivity, scalability, and service quality significantly, starting a greater degree of confidence between the former citizens and the local authorities.

Keywords: *AI-powered chatbots, citizen engagement, public service delivery, Indian municipalities, government services, digital transformation, operational efficiency.*

I.INTRODUCTION

Globally, the quick adoption of AI in public service delivery has changed how people interact with their governments and improved administrative effectiveness across many government agencies [1]. Innovative delivery channels in the form of municipal services are essential to the structure of 21st-century India, due to their account for a

significant portion of everyday citizen interactions, such as grievance redressal, queries, and complaint registration. But when paired with in-person meetings, phone conversations, and emails, traditional interaction techniques are data inefficient, leading to delays, excessive operating costs, and inconsistent service quality [2]. These antiquated practices create bottlenecks, irritate residents, and contributed an increasing strain on municipal workers. The study was prompted by the lack of interest in using an AI-driven chatbot to improve public service delivery, which might be simpler, more accessible, and resource-efficient. AI-driven chatbots that are backed by ML and NLP allow for automatic, real-time answers to citizen inquiries, substantially reducing down on wait times and improving service effectiveness overall[3]. These bots' easy integration with municipal databases and round-the-clock accessibility enhances the responsiveness and openness of government services [4]. The main goal of the research is to develop a chatbot system based on artificial intelligence that Indian municipal corporations can employ to encourage citizen participation and service delivery. To obtain information on the spot, offline individuals would probably engage with a chatbot that can connect to the backend municipal databases and automate responses to frequently asked questions, record complaints, provide status updates, and more. To make it easier for municipal employees to focus on more difficult instances that call for human judgment, the proposed system aims to reduce the amount of time that humans spend on

repetitive jobs [5]. Additionally, to improve its responsiveness to user interactions and comments, the chatbot will employ reinforcement learning. The proposed system will greatly improve accessibility, particularly for residents who lack the time, physical limitations, or geographic constraints to visit municipal offices. Furthermore, governments that automate citizen engagements stand to gain a great deal in terms of staff cost management and scalability for the growing urban population. The proposed system is an AI-enabled chatbot architecture that is scalable and combines real-time database access, dialogue management, and natural language processing to improve service delivery and close gaps in the public service domains. Additionally, it provides a comparison of the Chatbot system propose with conventional service models in terms of response time, resolution rate, and user happiness. The reinforcement learning (RL) model, a completely new way to adjust to improving public services' replies over time, also allows the chatbot to learn although with certain restrictions from how citizens interact with it. The technical specifics are also covered, including how these used BERT to categorize intent, sequence-to-sequence structures with attention (in transformer mode) for response generation, and Q-learning to gradually improve performance on Google Cloud. A deployment approach that makes use of containerized technologies, such as Docker and Kubernetes, to ensure the proposed system's scalability and high availability. In addition to addressing some of the most pressing issues in municipal service delivery, the research promotes the digital transformation of public administration more broadly and develops a useful manual on the use of AI as guidelines for future applications in government. The study is organized as follows: the second part examines the literature on AI-powered citizen engagement platforms and how these are used in public services. In Section III, the proposed chatbot system's architecture, NLP capabilities, reinforcement learning approach, and interface with local databases. Section IV presents the results and discussions that compare the chatbot's performance with standard practices (time dwell, solving efficiency, and user happiness). Finally, Section V concludes up the research by outlining the main conclusions and potential avenues for future study to enhance the provision of municipal services enabled by AI. In summary, the research demonstrates how AI-driven chatbots can significantly improve municipal service delivery in a way that is more

economical, accessible, and efficient. The proposed system streamlines citizen interactions via the computer interface by utilizing NLP, RL, and real-time database integration. Future research should focus on improving chatbot adaptability and expanding the use of AI to automate public functions.

II. RELATED WORK

Ndunagu et al. [6] examined data from the National Open University of Nigeria's E-ticketing system to assess an AI-driven chatbot utilizing a multinomial Naive Bayes algorithm for intent classification. The study demonstrated that the chatbot significantly diminished manual burden and optimized the management of student inquiries, hence improving communication and engagement in the Open and Distance Learning context. Chhibber et al. [7] demonstrated that AI transformed corporate processes by enhancing feedback management. Researchers created an intelligent chatbot utilizing Natural Language Processing to categorize consumer requests into 12 distinct classifications, attaining an accuracy of 91.35% with a Gradient Boosting Classification model. The study examined RF, SVM, K-NN, and Decision Tree algorithms, resulting in improved real-time review analysis and profit maximization for a paper manufacturing company. J. S et al. [8] built an AI-driven chatbot that responded to college inquiries and enhanced information accessibility in higher education. The system employed an NLP algorithm to assess student inquiries and archived responses to identify developing trends. It functioned incessantly, delivering polite and precise responses while handling multiple inquiries concurrently, therefore augmenting overall efficiency in college information services. A. D et al. [9] investigated the capacity of AI-driven chatbots to transform rural healthcare in India. In order to detect illnesses and offer initial medical advice, a symptom-checker chatbot that mimicked human-like interactions was developed. The study revealed that the chatbot optimized patient triage, decreased expenses, and improved access to vital medical information in underprivileged regions. Karnati et al. [10] designed BuddyBot, an AI-driven chatbot designed to improve English language acquisition through the utilization of sophisticated NLP models, such as GPT-2 and FLAN-T5. The study examined the difficulties faced by rural pupils with restricted access to quality resources. A five-

phase architecture was established, consisting of a user interface, pre-processing, a NLP stack, a language generating engine, and adaptive learning personalization. An E-Learn web-based platform offered interactive lessons supplemented by chatbot assistance. Validation studies indicated enhanced model performance via less training loss and elevated accuracy, so affirming the approach's effectiveness in personalizing education. Al-Mekhlal et al. [11] investigated the implementation of AI-driven bot solutions for digital transformation across multiple sectors, including healthcare, education, manufacturing, and energy. An EmailBot was created as a natural language processing tool to automate standard email interactions within a private cloud setting. The research contrasted an automated model with a conventional human-operated customer service system, emphasizing tasks including application installation, upgrades, link acquisition, permission requests, and status questions. Assessment metrics indicated significant enhancements in response time and improved accuracy in the automated model compared to the conventional support system. Jacob et al. [12] addressed the difficulties in obtaining timely and precise legal assistance within India's intricate legal framework. A sophisticated chatbot called LegalLink facilitated convenient access to vital legal information. The study utilized the Nomic AI text encoder to analyze user inquiries and the Retrieval-Augmented Generation model to get and incorporate external information into prompt templates. The suggested model produced contextually rich responses specific to legal questions. The study outlined the chatbot architecture, machine learning models, and employed methodology, thus transforming the accessibility of legal counsel for ordinary people. Christopherjames et al. [13] developed a health conversational bot employing NLP to aid users in understanding disease symptoms and treatment alternatives. The study tackled difficulties encountered by individuals without medical knowledge through the utilization of the Dialog flow API. The solution used across platforms including Google Assistant, Telegram, Slack, Messenger, websites, and mobile applications enabled users to submit text-based health inquiries and obtain precise medical suggestions. Database storage enabled enhanced patient analysis, while the solution aided in decreasing waiting times and optimizing resource utilization in healthcare service delivery. Singh, Joshi, and Domb [14] investigated the modernization and automation of healthcare

administrative functions and ongoing care operations through the integration of conversational AI. The research emphasized the substantial expenses and extended timeframes associated with in-person consultations, while investigating the capacity of AI chatbots and virtual assistants to improve efficiency and quality in healthcare. The study found significant issues include patient involvement, organizing unstructured data for automation, and enhancing CAI efficiency to increase business insights. Ganesh et al. [15] explored AI-driven automation to improve customer service responsiveness and efficiency. The research utilized ML and NLP to automate standard questions, enhance interactions, and minimize human error. A simulation performed over 100 epochs assessed various models, demonstrating enhanced training accuracy while indicating possible overfitting concerns. Methods like regularization, data augmentation, and cross-validation were suggested to enhance generalization. Model 2 exhibited the most equitable performance in terms of accuracy, recall, and F1 score, resulting in substantial cost reductions, increased customer satisfaction, and improved operational efficiency.

III. PROPOSED SYSTEM

The existing system of public service delivery and citizen participation in Indian municipalities is heavily reliant on traditional means such as mail, phone calls, and in-person visits. Despite being in use for many years, these techniques suffer from inefficiencies, redundancies, and accessibility issues. People must endure lengthy wait times on hold, ambiguous responses, and a lack of openness addressing their questions or grievances. Additionally, the requirement for human agents and shortened business hours leads to high operating costs and a significant barrier for people who live far from a government office. The goal of the proposed system is to deal with these issues by introducing AI-enabled chatbots that will revolutionize public service delivery. When NLP and ML are used, chatbots can handle a variety of tasks independently, including managing complaints, answering citizen questions, and exchanging information, among other services, while most of the time, human intervention is not nearly sufficient. Key Distinctions between the proposed System and the existing Configuration. The automation is the primary distinction between the existing setup and the proposed system. In

contrast to the traditional paradigm, which relies heavily on human agents, the AI chatbot will not only be accessible 24/7, but it will also deliver prompt responses to citizens. Simple and repetitive tasks like answering often-asked inquiries, providing service status updates, setting up appointments, and directing users to the appropriate department will be handled by it, reducing the workload of municipal staff and guaranteeing that problems can be fixed more rapidly. If the situation is more complicated, the chatbot can easily forward it to human agents so that no questions go unanswered. Because the system is designed to work with all digital platforms—websites, mobile apps, and social media channels—it may be accessed by a wider range of people, including those who have limited access to physical service places. The process of implementing the system begins with the chatbot's design and the creation of a robust knowledge base that makes use of the municipality's historical data, such as service lists, complaint forms, FAQs, etc. After processing the input, machine learning models enable the chatbot to comprehend and provide the user with a correct response. The proposed system understands and decodes the meaning of citizen inquiries by utilizing NLP algorithms, resulting in a more human-like connection. However, the chatbot can also learn from user chats and utilize reinforcement learning to improve its responses over time. During that stage, the proposed system will also be implemented on other digital platforms to reach and engage users. Following deployment, it must regularly monitor it, record its performance, get user input, and make necessary system improvements to ensure maximum user pleasure and optimal efficiency. The proposed AI-powered system provides a few benefits over the conventional configuration. It greatly decreases the time it takes for services to respond because prompt answers to citizen inquiries significantly enhance customer satisfaction and lessen impatience. Municipalities may serve many more inhabitants simultaneously and at the same quality of service due to the second component, which guarantees scalable deployment. By automating repetitive operations, the system can increase its human resources, freeing up human agents to work on more intricate, valuable problems. Additionally, because the chatbot offers 24/7 services to individuals regardless of their location or time zone, it also addresses a gap in accessibility. For any citizen who is unable to visit municipal offices during regular business hours due to work commitments, these is an excellent service. Because

the bot learns from the experiences it collected from previous conversations, AI and ML can also assist the system in improvisations over time. These contribute to increased accuracy and more individualized user experiences for the system. Additionally, the solution lowers operational expenses related to human data processing and call centers, which saves money. Overall, the proposed system will improve the efficiency, accessibility, and citizen-centricity of public service delivery in Indian towns, resulting in more satisfaction and confidence between the governed and the government. The below figure 1.1 shows the flowchart of proposed AI-Powered ChatBot system.

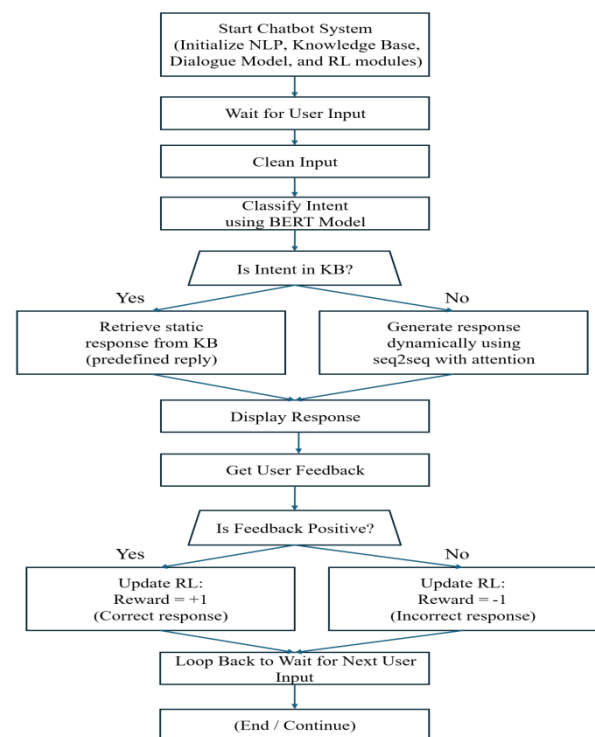


Fig 1.1 Flowchart of Proposed AI-Powered ChatBot System

System Architecture Design:

To propose an AI chatbot system that runs on the cloud and can be integrated with Chennai Greater Corporation portals (for property tax and water tax services) and Indian municipal websites such as TANGEDCO. NLP model deployment and real-time user interactions are effectively managed by the underlying architecture, which is housed on AWS Cloud services—Amazon SageMaker, Lambda, and API Gateway. For government databases (SQL or NoSQL), its database-agnostic design is supported by a RESTful API, WebSockets, and middleware

adapters. The cloud-first design achieves very low latency replies by facilitating smooth customer data access and returns from the user interface to the back and vice versa. The chatbot's performance is iteratively adjusted by a Q-learning-based feedback loop, which also makes public services scalable, responsive, and incredibly effective.

Preprocessing and NLP for Intent Recognition:

BERT, which is utilized here for robust intent recognition, powers the system's conversational component. The chatbot can effectively comprehend the various ways citizens could ask the same inquiry because of BERT's extensive contextual knowledge. Prior to being fine-tuned on municipal datasets such as the AI4Bharat-IndicNLP and INDO-ILSUM 2022 datasets, these undergo considerable pre-processing. To limit the quantity of the vocabulary, tokenization divides the input into smaller bits, and stemming lowers the words to their base words. Word weights which indicate whether a word is significant for a document or not and word embeddings which indicate the semantic linkages between the texts are employed by TF-IDF to generate feature vectors. Therefore, by increasing model accuracy through a thorough preprocessing pipeline, it produces thoughtful and precise responses that significantly improve public service engagement.

Dialogue Management and Response Generation:

The chatbot can manage conversations in a way that produces accurate responses due to the seq2seq (sequence-to-sequence) model with attention mechanism. Because it converts sequences (input user queries) to sequences (output chatbot responses), the seq2seq model is ideally suited for conversational AI. The proposed system can give more suitable responses by focusing on important components of the input due to the attentional process. After that, the model is trained using the past dialogue data which includes service inquiries and complaints to gain a solid grasp of how to react in various situations. During training, the model learns the mapping between inputs and expected outputs. It is then tuned to function well in a variety of conversational contexts, mimicking the delivery of public services.

ML for Continuous Improvement:

RL is used to gradually adjust the chatbot's responses based on input using the Q-learning algorithm. With user feedback, chatbot keeps getting better. Depending on whether a query was addressed

or escalated, end users score the bot's response fluency using the method. A correct response gains points, whereas a wrong response forfeits them. Having it in place enables the model to learn from every contact and enhance its performance. The Q-learning technique is dynamic because it can integrate simulated and real-time data from user interactions. Consequently, the chatbot learns to better address customer inquiries over time by adapting its behavior based on rewards and penalties for delivering high-quality service.

Integration with Backend Systems:

Using AWS Cloud for Simple Integration with Backend Systems: The chatbot integrates easily with municipal backend systems that AWS Cloud supports, allowing for real-time data processing, scalability, and efficiency. The API Gateway accepts the citizen request once it enters the chatbot via social media, mobile, or the web, and assigns processing of the request to an AWS Lambda function. After that, Lambda calls Amazon SageMaker, which classifies the input into the several intents trained BERT on. Next, it uses Seq2Seq with attention to construct the response. Lambda uses RESTful APIs to query the relevant municipal databases for real-time information, such as the complaint's status. After it receives the response, it is then transmitted back to the user interface via the API Gateway. Because resources are only spun up for use when needed, serverless computing and AWS auto-scaling enable minimal latency and cheap cost. To deliver seamless chatbot engagement for residents and consumers, cloud native event-driven architecture reduces infrastructure costs while maintaining high availability and responsiveness, which in turn increases citizen satisfaction.

Testing and Evaluation:

The chatbot undergoes rigorous testing after creation to ensure its functionality. It includes both automated and manual testing. Automated testing uses scripts that mimic hundreds of thousands of distinct user inquiries, whereas manual testing lets engineers mimic real-world user interactions to identify potential problem areas. Response accuracy, response time, and user satisfaction are among the performance-related parameters that are closely monitored by these tests. To evaluate the chatbot's performance with that of the same service using conventional service methods also employ A/B testing. The feedback obtained from these tests is

then utilized to improve the system's accuracy and ensure that it functions correctly in a variety of scenarios. To ensure that the proposed system is sound, dependable, and operational.

Deployment and Monitoring:

Following testing, the AI-infused chatbot is made available on websites, mobile applications, and social media platforms. In addition to supporting the deployment process with containerized technologies like Docker and Kubernetes, make sure that a broad spectrum of consumers will embrace the chatbot. These technologies provide the necessary flexibility and scalability to update and maintain the system without any downtime. When there is a system outage or performance deterioration, alerts are configured to automatically notify the development team. Additionally, to keep existing inputs based on evolving citizen demands, the chatbot is constantly retrained on new data for responses. In summary, the AI chatbot uses its NLP, dialogue management, and ongoing learning to deliver public services effectively. Real-time responsive exchanges are made possible by integration with back-end systems and a scalable cloud-based architecture for high availability. The chatbot, which is supported by thorough testing, deployment, and monitoring, automatically adjusts to enhance citizen contact and provides municipal services in a way that saves time and increases accessibility

IV. RESULTS AND DISCUSSION

The result and analysis will compare how well the AI-enabled chatbot system performs with the manual service delivery model that is still widely used in municipal services. Response time, query resolution time, customer happiness, and operational efficiency are the ultimate performance measures that are compared. The results were obtained by conducting real-world experiments with both systems and collecting data over time. These KPIs significantly improved with the AI-powered chatbot compared to the existing manual system.

Table 1 Response Time Comparison

Service Type	Mekhlal et al. 2023 [11] (Avg. Time)	Proposed System (Avg. Time)
General Inquiries	5–10 minutes	1–2 seconds
Complaint Registration	10–15 minutes	3–5 seconds
Service Status Query	15–20 minutes	2–4 seconds

The table's performance in terms of response times demonstrates the preceding significant improvement achieved by the proposed system over the existing system of Mekhlal et al. [11] dealing with general inquiries, which previously took 5–10 minutes, now takes only 1-2 seconds; logging into complaints, which previously took 10–15 minutes, now takes 3–5 seconds; and receiving information on service status, which used to take 15–20 minutes, now takes 2–4 seconds. Once more, the response time changes from seconds to milliseconds, demonstrating the system's ability to respond to users promptly.

Table 2 System Responsiveness and Adjustment Time (in seconds)

Service Type	Mekhlal et al. 2023 [11] (Resolution Rate)	Proposed System (Resolution Rate)
General Inquiries	80%	90%
Complaint Registration	75%	95%
Service Status Query	77%	85%

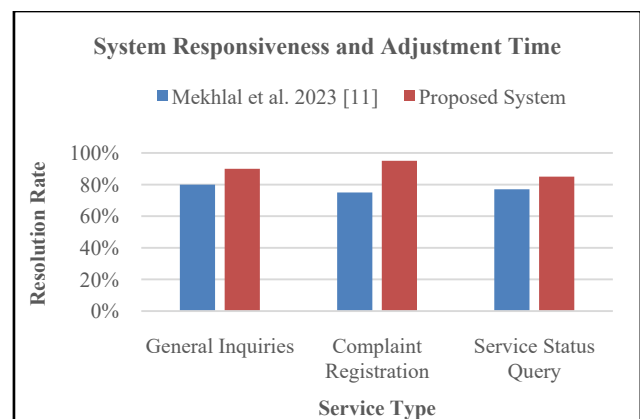


Fig 1.2 Graphical Illustration of System Responsiveness and Adjustment Time in Proposed Vs Existing System.

The table resolution rates of the proposed system and the existing system of Mekhlal et al. [11] are presented. Significant improvements on a wide range of requires are demonstrated by the proposed system. The cost of answering general questions is 90% as opposed to the existing rate of 80%. Complaint registration has improved the most, going from 75% to 95%. Additionally, the percentage of service status query resolutions rises from 77% to 85%. The results in Fig 1.2 demonstrates that the proposed system not only speeds up request response times but also increases task completion

rates and accuracy while satisfying user requests.

Table 3 User Satisfaction Comparison

Service Type	Mekhlal et al. 2023 [11] (Satisfaction Score)	Proposed System (Satisfaction Score)
General Inquiries	3.2/5	4.7/5
Complaint Registration	3.5/5	4.8/5
Service Status Query	3.0/5	4.6/5

User satisfaction with the proposed system has improved over the existing system of Mekhlal et al. [11], as seen in the above table. A significant improvement over the prior user experience, the user satisfaction rate for general questions has increased from 3.2/5 to 4.7/5. Service status queries have increased from 3.0/5 to 4.6/5, and complaint registration satisfaction has increased from 3.5/5 to 4.8/5. The customers believe that the proposed system is more dependable, efficient, and user-friendly than the existing one, as evidenced by the improved satisfaction ratings. Chatbots Driven by AI Revolutionize Public Service Provision in Indian municipalities, like those in other international economies, have begun deploying AI-powered chatbots to increase response efficiency, resolution rate, and user happiness in general. People receive prompt responses to all inquiries, complaints, and service status inquiries since the system cuts down on waiting times. The model change allows for a speed and efficiency that surpasses the delays and irregularities that are common in traditional methods. The chatbot offers round-the-clock accessibility, smooth connections to local databases, and personalized user interaction. Its ability to handle a large number of public interactions simultaneously reduces the workload for municipal employees, allowing them to concentrate on more complex tasks. When paired with backend systems for real-time updates, the chatbot increases the accountability and openness of government services. Additionally, NLP and RL enable the chatbot to learn the best response style based on user behavior, allowing it to take advantage of continuous improvement with changing responses. Applications can be expanded beyond municipal services to include industries like healthcare, education, and transportation, where automated assistance can promote citizen participation. By reducing operating costs and improving accessibility, the chatbot helps

to rebuild public confidence in the government and fosters scalable AI-driven governance models that facilitate better service delivery at the national level.

V.CONCLUSION

In conclusion, the AI-enabled chatbot system greatly increases citizen satisfaction while also revolutionizing public service delivery in Indian towns by reducing response times and increasing resolution rates. It uses NLP and RL together with its backend system integration, which gives it access to real-time data, to continuously improve. The system has its own restrictions, of course. First, regional dialects and mixed language inputs may impact how accurately NLP models respond. Second, even though the majority of the work is automated and requires fewer human processes, some resources and problems cannot be solved without human intervention, which makes complete automation impossible. Third, the deployment of chatbots necessitates high upfront expenditures and advanced technological skills, which is an obstacle for local governments that do not have the resources on hand. Future improvements could further optimize the system. Trimming NLP models with local language datasets to increase answer generation accuracy is one area that could use improvement. Furthermore, by incorporating sentiment movement, the chatbot can discern a citizen's emotions and escalate the issue as necessary. Lastly, the chatbot should have features that facilitate voice-based interactions to increase accessibility for individuals with low levels of digital literacy. These will strengthen AI-powered citizen participation, increasing the effectiveness, transparency, and inclusivity of public services. If some of the drawbacks are resolved and certain future improvements are made, AI chatbots have the potential to become a fundamental component of digital government in India and foster responsiveness and trust in municipal services.

REFERENCES

- [1]. C. Waladi, M. S. Lamarti, and M. Khaldi, "Transforming Customer Experiences with Customer-Centric AI-Powered Chatbots in the Fourth Industrial Revolution," *Mediterranean Smart Cities Conference (MSCC)*, pp. 1–5, May 2024, doi: 10.1109/mscc62288.2024.10697060.
- [2]. T. Mathur and R. Kumar V. V., "AI-Powered Chatbots in Health Insurance: Assessing Their Effectiveness in Consumer Search Processes," *11th International Conference on Electrical Engineering, Computer Science and Informatics (EECSI)*, pp. 579–582, Sep. 2024, doi: 10.1109/eeesi63442.2024.10776387.
- [3]. A. Bhharathee, S. Vemuri, B. Bhavana, and K. Nishitha, "AI-

- Powered Student Assistance Chatbot,” International Conference on Intelligent Data Communication Technologies and Internet of Things (IDCIoT), Jan. 2023, doi: 10.1109/idciot56793.2023.10053439.
- [4]. L. S. Saoud and A. A. Romman, “AI-Powered Chatbots in the telecommunications industry in the UAE: Studying the factors impacting consumer adoption,” 2022 13th International Conference on Information and Communication Technology Convergence (ICTC), pp. 1549–1553, Oct. 2024, doi: 10.1109/ictc62082.2024.10827670.
 - [5]. X. Kong, G. Wang, and A. Nichol, *Conversational AI with Rasa: Build, test, and deploy AI-powered, enterprise-grade virtual assistants and chatbots*. Packt Publishing Ltd, 2021.
 - [6]. J. N. Ndunagu, R. G. Jimoh, U. Chidiebere, and G. Deborah. Opeoluwa, “Enhanced Open and Distance Learning Using an Artificial Intelligence (AI)-Powered Chatbot: a Conceptual Framework,” 5th Information Technology for Education and Development (ITED), pp. 1–4, Nov. 2022, doi: 10.1109/ited56637.2022.10051575.
 - [7]. M. Chhibber, R. Mittal, and A. Singh, “AI-powered Chatbot for Improved Customer Feedback Management using NLP,” 2022 13th International Conference on Computing Communication and Networking Technologies (ICCCNT), Jul. 2023, doi: 10.1109/iccant56998.2023.10308364.
 - [8]. J. S. J. S, S. M, and G. R. Hemalakshmi, “College Enquiry Chatbot,” 2nd International Conference on Vision Towards Emerging Trends in Communication and Networking Technologies (ViTECoN), pp. 1–4, May 2023, doi: 10.1109/vitecon58111.2023.10156978.
 - [9]. A. D. M. M, P. T, T. M, P. S. S. A, and N. E. R, “Revolutionizing rural healthcare in India: AI-Powered chatbots for affordable symptom analysis and medical guidance,” 2022 International Conference on Inventive Computation Technologies (ICICT), Apr. 2024, doi: 10.1109/iciet60155.2024.10544758.
 - [10]. D. J. P and S. B. Karnati, “BuddyBot: AI Powered Chatbot for Enhancing English Language Learning,” IEEE International Conference on Interdisciplinary Approaches in Technology and Management for Social Innovation (IATMSI), pp. 1–6, Mar. 2024, doi: 10.1109/iatmsi60426.2024.10502595.
 - [11]. .M. Al-Mekhlal, M. Al-Buraik, and M. Al-Lubli, “Digital Transformation: AI-Powered Bot Solutions and Automation for Customer Services,” International Conference on Digital Applications, Transformation & Economy (ICDATE), pp. 1–7, Jul. 2023, doi: 10.1109/icdate58146.2023.10248458.
 - [12]. S. Jacob, P. M. Jacob, J. Cheriyan, and S. Ismail, “Legal Assistance Redefined: Transforming Legal Access with AI-Powered LegalLink,” 2022 IEEE International Conference on Signal Processing, Informatics, Communication and Energy Systems (SPICES), pp. 1–6, Sep. 2024, doi: 10.1109/spices62143.2024.10779909.
 - [13]. J. E. Christopherjameset al., “Natural Language Processing based Human Assistive Health Conversational Agent for Multi-Users,” 2021 Second International Conference on Electronics and Sustainable Communication Systems (ICESC), pp. 1414–1420, Aug. 2021, doi: 10.1109/icesc51422.2021.9532913.
 - [14]. A. Singh, S. Joshi, and M. Domb, “Embedded Conversational AI, Chatbots, and NLP to Improve Healthcare Administration and Practices,” 2nd International Conference on Automation, Computing and Renewable Systems (ICACRS), vol. 7, pp. 38–45, Dec. 2023, doi: 10.1109/icacrs58579.2023.10404985.
 - [15]. C. N. Ganesh, M. C. Devi, B. R. Celia, G. Shrivastava, V. Vijayalakshmi, and S. P, “Enhancing Customer Service Response and Efficiency with AI-Driven Automation,” 4th International Conference on Technological Advancements in Computational Sciences (ICTACS), pp. 1787–1793, Nov. 2024, doi: 10.1109/ictacs62700.2024.10840551.